Exponential mistakes and polynomial guarantees for circuit Occam learning

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Abstract

An online circuit Occam learner predicts using a smallest circuit consistent with all previously revealed labels. Does a small target circuit imply few mistakes when inputs arrive from changing or adversarial regions? The answer depends decisively on how the learner selects among equally small circuits.

For every deterministic selection rule based on the previous history, the conjunction of d variables admits a distinct-input ordering with at least $2^d - d - 2$ mistakes. We certify exact circuit minimality, including unrestricted NAND circuits, where the target's minimum gate count is $2d - 2$. This gives an exponential lower bound in the target gate count. The corresponding two-input-gate result was already proved in earlier work; we present a short proof and a NAND certificate argument.

Conversely, freshly sampling uniformly among the smallest consistent circuits gives $O(s^2 \log(d+s))$ total expected mistakes for a target of size s, on every nonanticipating adaptive input sequence. Majority prediction within the minimum-size class has a deterministic bound of the same order. For any history-based tie rule, arbitrary adaptive interleaving of K fixed input environments gives polynomial mistake bounds when the environments have controlled disagreement probabilities; in particular, arbitrary uniform subsets of the Boolean cube give $O(K(s+1)\log(d+s+2)(d+1))$. Local random shuffling of the conjunction example reduces exponential mistakes to quadratic expected mistakes. We also prove bounds for continuing adaptive distribution drift and a precise constant-fraction-elimination assumption. Exponentially small distance from uniform, even with exponentially small total successive drift, does not alone prevent exponentially many mistakes. All results concern noiseless online feedback, not unlabeled transfer after training.

1 The question and the prediction protocol

Simplicity-based prediction can extrapolate because one compact rule may apply in many regions of its input space. A worst-case online guarantee asks for more: after every prediction the true label is revealed, and the total number of mistakes should be polynomial in the complexity of that rule, even if the queried region changes. We study this question for exact minimum-circuit prediction, with unlimited computation.

Fix $d \geq 1$ and a target $f : \{0,1\}^d \to \{0,1\}$. At round t, an input x_t is presented, the learner predicts, and $f(x_t)$ is revealed. The label function never changes. Inputs may repeat, but exact consistency makes a repeated input harmless. The conjunction-ladder lower bounds use no repetitions; the near-uniform adaptive construction later uses repeats. The mistake count is over the whole sequence, including an infinite sequence where indicated. This is an online guarantee with feedback, not a fixed-model test guarantee under an arbitrary new distribution.

A circuit is an acyclic list of gates with two initial constant wires and the d variable wires. Each gate reads earlier wires; fan-out and sharing are unrestricted. The output is the final gate, and

size is the number of gates. We use either all sixteen two-input Boolean gates or only two-input NAND gates, stating the basis explicitly. Binary AND/OR and unary NOT are also covered by the first lower-bound proof. No free inversions are allowed in the NAND-only result. The target size $s = C(f)$ is the minimum gate count in the chosen basis. All logarithms are natural unless a base is displayed. Every confidence parameter δ below lies in $(0, 1)$.

Three prediction rules must be distinguished:

1. A *deterministic history-based selector* chooses one smallest consistent circuit from previous observations and evaluates that circuit on the next input. This includes a fixed ordering of circuit descriptions to resolve ties. The circuit choice does not use the current input.

2. A *fresh uniform selector* draws independently and uniformly from the currently smallest consistent circuit descriptions each round. The current input is fixed before that fresh draw, which is private until then. An adaptive input chooser may use all previous predictions.

3. A *minimum-class majority predictor* takes the majority label of the smallest consistent descriptions on the current input. It may equivalently select a smallest circuit giving that label, but its circuit selection then uses the current input.

Uniformity over descriptions need not mean uniformity over functions. Our general counting bounds allow duplicate descriptions. None of these rules is automatically the posterior of a Gaussian neural-network prior.

The main distinction. The first rule has an exponential worst-case lower bound, even for an AND target. The second and third have polynomial upper bounds for every allowed input order. Thus "choose a smallest consistent circuit" alone does not specify a learner well enough to determine its online behavior. The exponential construction can use a simply described order, so low description complexity of the input sequence alone is not a sufficient positive assumption. Our positive results for the first rule instead give explicit stochastic conditions on how new inputs are selected.

Circuit counts. There are exactly

$$N_j = \prod_{i=1}^{j} 16(d + i + 1)^2$$

size-j descriptions in the all-binary-gate model, and the same product without the factors 16 for NAND. In particular, with $N_0 = 0$,

$$h_s := \log \sum_{j=1}^{s} N_j \le \log s + s \log 16 + 2s \log(d + s + 1), \tag{1}$$

$$B_s := s + \sum_{j=1}^{s} \log N_j \le s + s(s + 1) \log(4(d + s + 1)). \tag{2}$$

The learner need not know s; these are proof budgets. The realizing target ensures that no consistent minimum-size choice ever exceeds s.

Relation to earlier work. The conjunction ladder appeared in the Vaintrob–Hänni slides [1]. Hänni and Claude [2] proved its exact minimum-circuit version for all binary gates and distinguished it from Bayesian averaging. We reproduce that result with an elementary support proof, then give a separate NAND-only proof. The positive arguments are self-contained finite-class and version-space calculations, including the standard halving principle [3]. They make the input, tie-breaking, and complexity conventions explicit. These are conventional proofs with independent AI-agent checks, not Lean formalizations.

2 An exponential lower bound for exact circuit minimization

The conjunction ladder below is the construction from Appendix K of the Vaintrob–Hänni slides [1], proved for unrestricted binary-gate circuits by Hänni and Claude [2]. We give an elementary support-based proof and make the prediction protocol explicit. Its lower bound is exponential in the target's *minimum gate count*, and every hypothesis used by the learner really is a globally smallest consistent circuit.

Throughout this section a circuit is a finite, acyclic, single-output circuit over either all sixteen two-input Boolean gates or the standard basis consisting of binary AND and OR and unary NOT. Constants are available as source wires, and arbitrary fan-out is allowed. The output must be a gate, so a circuit has at least one gate. Size counts every gate, including NOT, and $C(f)$ denotes the minimum size of a circuit computing f on the full Boolean cube.

A *history-only minimum-circuit learner* chooses a circuit h_t of minimum gate count among all circuits consistent with the first $t-1$ labeled examples, and predicts $h_t(x_t)$. Its choice of h_t can depend on the entire past, but not on the current query x_t. For deterministic learners this includes every fixed ordering of equal-size circuit descriptions or equal-complexity functions.

Lemma 1 (Support forces exact minimality)**.** *A circuit of s gates depends on at most $s+1$ input variables. Suppose $h(1^d) = 1$ and $h(x) = 0$ whenever $|x| < k$. Then h depends on at least k variables. If it depends on exactly k variables, it is the conjunction of those variables.*

Proof. Prune gates having no path to the output, leaving $t \le s$ gates. Of their at most $2t$ input slots, at least $t-1$ reference other remaining gates. Thus at most $t+1 \le s+1$ slots can hold distinct input variables. This argument permits unary gates and shared wires.

Let S be the essential-variable set of h, and let 1_S denote its indicator in $\{0,1\}^d$. If $|S| < k$, then $h(1_S) = h(1^d) = 1$, contradicting the lower-weight negative labels. If $|S| = k$, extend any assignment on S other than 1_S by zeros outside S. The resulting input has weight below k, so its label is zero. The all-ones assignment on S has label one. Hence $h(x) = \bigwedge_{j \in S} x_j$. □

Theorem 2 (Conjunction ladder for exact minimum circuits)**.** *Let $d \ge 3$ and $f(x) = \bigwedge_{j=1}^{d} x_j$. Then $C(f) = d-1$. For every deterministic history-only minimum-circuit learner, there is a permutation of $\{0,1\}^d$ on which its total number of mistakes is at least*

$$\sum_{k=2}^{d-1} \binom{d}{k} = 2^d - d - 2 = 2^{C(f)+1} - C(f) - 3.$$

The permutation may depend on the learner's tie rule.

Proof. First reveal 1^d with label one, then 0^d and all singleton inputs with label zero. For $k = 2, \dots, d-1$, process the inputs of weight k, choosing the order within the level as follows.

At the beginning of this phase, every input of weight less than k has been revealed with label zero. At an intermediate step, let $\mathcal{R}$ be the collection of k-sets whose indicator inputs have not yet been queried. If $\mathcal{R} \neq \varnothing$, each

$$h_S(x) = \bigwedge_{j \in S} x_j, \qquad S \in \mathcal{R},$$

is consistent and has a circuit of $k-1$ gates. Every consistent circuit has at least $k-1$ gates by Lemma 1. Consequently the *global* minimum over all consistent circuits is exactly $k-1$.

Every minimizing circuit depends on at most k variables and at least k variables. The same lemma therefore says that it computes h_S for a k-set S. Consistency forces $S \in \mathcal{R}$. Query 1_S. The learner predicts one and the target label is zero. Moreover, for another k-set T,

$$h_T(1_S) = 1 \quad \Longleftrightarrow \quad T \subseteq S \quad \Longleftrightarrow \quad T = S.$$

Thus this query eliminates precisely one of the surviving k-conjunctions. Repeat until the level is exhausted. Every input in every level $k = 2, \ldots, d-1$ is a mistake.

The query can be determined before prediction because the learner's current hypothesis is a deterministic function of its past. Equivalently, simulate the construction in advance to obtain a fixed permutation for this learner. Finally, f depends on all d variables, so Lemma 1 gives $C(f) \geq d-1$; the usual AND tree attains equality. □

A fixed tie rule and a fixed input order. There is also a direct nonadaptive example. At each size $s \geq 1$, rank canonical left-associated conjunction circuits on $(s+1)$-sets first, ordered lexicographically by their supports, and rank all other circuits afterward in any fixed order. Choose the first consistent circuit of minimum size. Present 1^d, 0^d, all singletons, and then the levels of weight $2, \ldots, d-1$ in that same lexicographic order. At every step after the singleton level, the current hypothesis is the conjunction indexed by the next input. The number of mistakes is exactly $2^d - d - 2$. Both orderings are fixed independently of the run. The theorem is stronger in a different direction: every deterministic history-only tie rule has some bad ordering, without asserting one ordering bad for all tie rules.

Why the singleton level is omitted. Under the mandatory-output-gate convention, a one-gate $x_i \wedge x_j$ is consistent with the initial positive example, the zero vector, and every singleton negative. Therefore the first level need not incur any mistakes. If instead an input or constant wire may itself be a zero-gate output, the proof starts at $k = 1$ and yields $2^d - 2$ mistakes after the two initial examples, with the same target complexity $d-1$. This stronger count uses a different, explicitly stated circuit convention.

The complexity measure matters. The theorem is literally exponential in the minimum number of gates. For the explicit gate-list language that encodes size in unary and then gate types and wire addresses, the target has description length $b = \Theta(d \log d)$. In that measure the conclusion is $2^{\Omega(b/\log b)}$, not a proved $2^{\Omega(b)}$ bound. A constant-factor compilation into a NAND-only basis does not automatically transfer the theorem: compilation need not preserve which consistent circuits are smallest. The next section proves the NAND-only statement directly.

Scope of the tie-breaking quantifier. The deterministic theorem does not permit an adversary to inspect a hidden fresh random circuit before selecting its query. Nor does it cover a deterministic learner that uses the current query to choose between minimum circuits. For example, after receiving x_t, a learner may take the majority label among all current minimum circuits and then select a

minimum circuit giving that label. This still selects a smallest consistent circuit at every round. Each mistake halves the candidates at the current size, or exhausts that size. If N_s bounds the number of size-s descriptions, its mistakes are at most

$$\sum_{s=1}^{C(f)} \left(1 + \log_2 N_s\right) = O\Big(C(f)^2 \log(d + C(f))\Big).$$

Thus specifying only "a smallest circuit each round" leaves a material ambiguity: the exponential lower bound concerns a single deterministic hypothesis chosen from the past, before using the current query.

3 The same separation for pure NAND circuits

The separation also holds when every gate is a two-input NAND. Size means the number of NAND gates; intermediate results and input wires may be reused without restriction. Constants may be allowed or omitted. Negations and complemented edges are not free operations in this model.

Lemma 3 (A positive certificate for a NAND circuit)**.** *Let a NAND circuit with $s \geq 1$ gates compute h, and suppose $h(1^d) = 1$. There is a set $R \subseteq [d]$ with $|R| \leq \lfloor s/2 \rfloor + 1$ such that $h(x) = 1$ whenever $x_i = 1$ for every $i \in R$. In particular, $h(1_R) = 1$.*

Proof. Evaluate every gate at 1^d, calling it true or false according to its value. Starting at the true output gate, retain one false-valued input of each selected true NAND gate, and both true-valued inputs of each selected false NAND gate. Recurse to the selected sources, stopping at variables and constants. Identify repeated occurrences of the same source or gate, but keep repeated edges when both input slots select the same source. The resulting proof graph is a connected rooted directed acyclic multigraph.

Let T, F, L be its numbers of true gates, false gates, and distinct source leaves, respectively. Every selected false gate has a selected true parent, whereas every true gate selects only one input. Thus $F \leq T$: choose one true parent of each false gate; distinct false gates cannot choose the same parent. There are $T + 2F$ selected edges and $T + F + L$ vertices. Connectedness gives

$$T + F + L - 1 \leq T + 2F, \qquad L \leq F + 1.$$

Since $T + F \leq s$ and $F \leq T$, we obtain $L \leq \lfloor s/2 \rfloor + 1$.

Let R consist of the distinct variable leaves, so $|R| \leq L$. Every one of these variables is one at 1^d. Fixing them to one forces every selected gate to retain its value at 1^d, by induction in circuit order: a selected false NAND has both inputs forced true, and a selected true NAND has one input forced false. Constants retain their values without any restriction on the input. The output therefore remains one. The argument explicitly permits sharing and repeated input wires. □

Corollary 4 (Exact NAND complexity of conjunctions)**.** *For $k \geq 2$, the conjunction of k distinct variables has minimum NAND-circuit size $2k - 2$.*

Proof. An AND of two signals is a NAND followed by a NAND with its two inputs tied together. Replacing every gate in an AND tree gives $2k - 2$ gates. Conversely, every positive certificate for a conjunction must include all its k variables. The lemma gives $k \leq \lfloor s/2 \rfloor + 1$, hence $s \geq 2k - 2$. □

Theorem 5 (Exponential mistakes for smallest NAND circuits)**.** *Fix $d \geq 3$ and let $f(x) = \bigwedge_{i=1}^{d} x_i$. Consider any deterministic learner which, before receiving the next input, chooses a minimum-size*

NAND circuit consistent with all preceding examples. There is a permutation of $\{0,1\}^d$, labeled throughout by f, on which the learner makes at least

$$2^d - d - 2 = 2^{s/2+1} - s/2 - 3 \quad \textit{mistakes}, \qquad s = C_{\mathrm{NAND}}(f) = 2d - 2.$$

The permutation may depend on the deterministic learner, but can be fixed before the sequence is presented.

Proof. First present 1^d with label one, then every input of weight zero or one with label zero. For $k = 2, \ldots, d-1$, present the weight-k inputs as follows.

At any point in stage k, every input of weight below k has already been labeled zero. If a consistent circuit has r gates, the lemma provides a positive certificate R with $|R| \le \lfloor r/2 \rfloor + 1$. Consistency requires $|R| \ge k$, since otherwise $h(1_R) = 1$ contradicts a previous label. Therefore every consistent circuit has at least $2k-2$ gates.

As long as some weight-k input 1_S remains unshown, the conjunction $\bigwedge_{i \in S} x_i$ is consistent: it is one at 1^d, zero below weight k, and among weight-k inputs it is one only at 1_S. It has $2k-2$ NAND gates. Thus the learner's minimum consistent size is exactly $2k-2$. A positive certificate R for its chosen circuit has $|R| \le k$, and consistency forces $|R| \ge k$. Present 1_R. This input has not been shown before, the learner predicts one, and the target label is zero because $|R| = k < d$.

Repeating this step presents every weight-k input as a mistake. Summing over $k = 2, \ldots, d-1$ gives $\sum_{k=2}^{d-1} \binom{d}{k} = 2^d - d - 2$. The initial points and these levels comprise the whole hypercube. Because the learner is deterministic, the constructed sequence can be simulated and fixed in advance. The target labels never depend on the learner. □

A simple fixed tie rule and input order. For every $k \ge 2$, fix the usual left-associated AND chain on a sorted list of k variable indices and replace each AND by its two-NAND implementation. At size $2k-2$, place these canonical circuits first, ordered lexicographically by their supports, and place all remaining circuit descriptions afterward in any fixed order. At other sizes, use any fixed order. Present 1^d, then 0^d, then the remaining inputs in increasing Hamming weight, using lexicographic support order within each level. At every stage $k \ge 2$, the minimum consistent size is $2k-2$, and the first surviving canonical conjunction has exactly the support of the next input. This fixed tie rule and fixed input order therefore realize the lower bound. No classification of all minimum-size NAND hypotheses as conjunctions is required.

Conventions. The argument starts at $k = 2$, so it applies both when outputs must be gates and when direct wires or constants are allowed as zero-gate outputs. If desired, the certificate lemma extends to those zero-gate cases: a projection has a one-variable certificate and constant one has the empty certificate. The result concerns gate count. In a literal gate-list code with input addresses, the target's description length is $\Theta(d \log d)$, so the lower bound becomes $2^{\Omega(\ell/\log \ell)}$ in that description length.

4 Polynomial mistakes for fresh uniform ties and minimum-class majority

Let $\mathcal{C}_j$ be a finite set of circuit descriptions of size j, and write $N_j = |\mathcal{C}_j|$. Distinct descriptions computing the same function are counted separately. Let f have a description of size s. All labels are noiseless, every label is revealed after prediction, and all subsequent predictions must be consistent with every revealed label. No algorithmic efficiency is asserted.

4.1 Fresh uniform tie-breaking: arbitrary adaptive input order

Theorem 6 (Fresh uniform minimum-description prediction)**.** *On each round, choose a fresh uniformly random description from the currently consistent descriptions of minimum size, and predict with it. The input may be chosen adaptively using the entire past and the learner's previous predictions, but must be chosen before the fresh random choice of the current description. Put*

$$B_s = s + \sum_{j=0}^{s} \log \max\{1, N_j\}.$$

Then the total number M_∞ of mistakes over an arbitrarily long, possibly infinite, input sequence satisfies

$$\mathbb{E}M_\infty \le B_s, \qquad \mathbb{P}\{M_\infty \ge (e-1)B_s + \log(1/\delta)\} \le \delta.$$

For the all-binary-gate convention of this paper, and hence also for NAND,

$$B_s \le s + s(s+1)\log(4(d+s+1)),$$

as in (2). The empty size-zero class is omitted. In particular this is a polynomial total-mistake guarantee for exact minimum-size prediction under fully adaptive nonanticipating input order. The learner need not know s.

Proof. Immediately before a label is revealed, let j be the minimum surviving size, let n be the number of surviving descriptions at that size, and let q be the fraction of these descriptions wrong on the current input. The conditional probability of the learner's mistake is exactly q. If $q < 1$, the surviving count at this same size becomes $n(1-q)$ and

$$q \le -\log(1-q) = \log n - \log(n(1-q)).$$

Thus the sum of all such q while size j is the minimum is at most $\log N_j$. If $q = 1$, the size-j class is exhausted and this transition costs one. Minimum size never decreases. It never exceeds s, and the size-s class cannot be exhausted because the target survives. There are therefore at most s exhausted tiers. On every realization,

$$\sum_{t \ge 1} q_t \le B_s.$$

For the actual mistake indicators Z_t, conditional on the past and the current input, $\mathbb{E}Z_t = q_t$ and

$$\mathbb{E}\exp\{Z_t - (e-1)q_t\} = e^{-(e-1)q_t}(1 + (e-1)q_t) \le 1.$$

Consequently $\exp\{M_T - (e-1)\sum_{t \le T} q_t\}$ is a nonnegative supermartingale, also when the next input is chosen from the full past. Monotone convergence proves the expectation bound, and Ville's inequality proves the time-uniform tail bound. The counting estimate is (2). □

Remark 7 (What the tie rule does and does not say)**.** *Uniformity is over the specified finite set of minimum-size descriptions, not automatically over distinct Boolean functions. Uniform sampling over the latter also works if the corresponding finite counts are used. The random choice must be fresh and private until the current input is fixed. A single random ordering of descriptions, reused forever, is not conditionally uniform after an adaptive input chooser has observed predictions, so the above proof does not apply to that rule.*

Corollary 8 (Minimum-tier majority)**.** *Predicting the majority label among the currently consistent descriptions of minimum size, with arbitrary ties, makes at most $2B_s$ mistakes on every input sequence. This can also be implemented as a query-dependent choice of a minimum-size consistent circuit agreeing with that majority.*

Proof. The same pathwise bound on $\sum q_t$ holds without any random prediction rule. Each majority mistake has $q_t \geq 1/2$. □

4.2 The conjunction ladder under fresh uniform ties

Proposition 9 (Fresh uniform ties on the same ladder)**.** *Use the all-binary-gate or AND/OR/NOT basis, with all gates charged. Suppose that before every prediction the learner freshly samples uniformly from all consistent circuits of minimum gate count. Consider the level-by-level input order from Section 2, with the order within each level chosen before the current fresh random draw. For each $k = 2, \ldots, d-1$, its expected mistakes in level k are exactly*

$$H_{\binom{d}{k}} = \sum_{r=1}^{\binom{d}{k}} \frac{1}{r}.$$

Thus its expected mistakes in these levels are

$$\sum_{k=2}^{d-1} H_{\binom{d}{k}} = \Theta(d^2),$$

and the initial $d+2$ queries contribute at most $d+2$ further mistakes.

Proof. Lemma 1 shows that the minimum circuits in level k compute precisely the surviving k-conjunctions. Permuting input coordinates is a bijection between the size-$(k-1)$ circuits computing any two such conjunctions. They therefore have equal numbers of minimum-size descriptions, in each of these two bases. Since all descriptions computing a given function agree on the entire observed history, uniformity over minimum circuit descriptions induces uniformity over the surviving conjunctions.

If r remain, any unqueried input of weight k is predicted incorrectly by exactly one of them. The conditional mistake probability is $1/r$, and the revealed negative label removes exactly that conjunction. Summing gives the harmonic number. For the asymptotic estimate, $H_n \leq 1 + \ln n$ gives an upper bound $O(d^2)$. For $\Theta(d)$ central values of k, $\binom{d}{k} = \exp(\Omega(d))$, so $H_{\binom{d}{k}} \geq \ln \binom{d}{k} = \Omega(d)$, proving the lower bound. □

5 Arbitrary history-dependent tie-breaking in changing environments

The next results apply to every finite class H containing f, with $h = \log |H|$. On each round the learner selects a consistent member of H using the previous history, before seeing the next sampled input. It may randomize and its tie rule may depend on all previous source identities, inputs, labels, and predictions. Minimum-size consistent circuits satisfy this requirement with $H = \bigcup_{j \leq s} \mathcal{C}_j$, provided the choice of a minimum circuit does not depend on the current sampled input. In this case

$$h \leq \log s + s \log 16 + 2s \log(d + s + 1).$$

The input-independent choice condition matters: a different consistent hypothesis chosen after seeing each input can have much greater aggregate error than any individual consistent hypothesis.

Theorem 10 (Adaptively interleaved random-order environments)**.** *Fix K finite input pools of sizes $m_1, \dots, m_K \geq 1$; repeated inputs inside or between pools are allowed. Independently uniformly permute each pool. At each round a scheduler, using the full observed past but no unrevealed entries, chooses a nonexhausted pool and reveals its next entry. It may stop at any time. Then every learner just specified satisfies*

$$\mathbb{E}M \leq K + 2(h+1)\sum_{i=1}^{K} H_{m_i-1}, \qquad H_0 = 0.$$

For $0 < \delta < 1$, put

$$a = h + \log\frac{2\sum_i m_i}{\delta}, \qquad B_\delta = K + 2a\sum_i H_{m_i-1}.$$

With probability at least $1-\delta$,

$$M \leq (e-1)B_\delta + \log(2/\delta).$$

In particular, if each pool has at most 2^d entries, expectation is $O(K(h+1)(d+1))$, and high probability adds only polynomial dependence on $\log(K/\delta)$ and d.

Proof. For pool i, let $R_{i,n}$ be the maximum pool error fraction of a member of H agreeing with the first n entries in that pool, ignoring all other labels. A candidate with error fraction r survives that prefix with probability at most $(1-r)^n \leq e^{-nr}$. Thus

$$\mathbb{P}(R_{i,n} > u) \leq \min\{1, e^{h-nu}\}, \qquad \mathbb{E}R_{i,n} \leq (h+1)/n.$$

At the next visit to pool i, the selected predictor is consistent with that prefix; all of its pool errors lie in the $m_i - n$ remaining entries. Its conditional error probability is at most $m_i R_{i,n}/(m_i - n)$. Charging each used local count (i,n) once and enlarging to all possible local counts gives the expectation bound because

$$\sum_{n=1}^{m_i-1} \frac{m_i}{n(m_i-n)} = 2H_{m_i-1}.$$

The first visit to each pool costs at most one. This argument is unchanged by arbitrary adaptive interleaving because each unused next entry remains uniformly distributed among its own pool's remaining entries.

For high probability, a union bound makes all inequalities $R_{i,n} \leq a/n$, $1 \leq n < m_i$, hold except with probability $\delta/2$. Stop the mistake process immediately before using a failed prefix. Until that stopping time its total conditional error probabilities are at most B_δ. The exponential supermartingale used above bounds its mistakes by $(e-1)B_\delta + \log(2/\delta)$ except with probability $\delta/2$. On the simultaneous good-prefix event the process never stops, proving the claim. □

Theorem 11 (Interleaved iid environments)**.** *Fix K distributions $D_1, \dots, D_K$. A scheduler uses the full past to choose a source, then receives a fresh independent input from that source. For every source having at least one nonzero disagreement risk in H, let*

$$\varepsilon_i = \min\{D_i(g \neq f) : g \in H,\ D_i(g \neq f) > 0\}.$$

Omit and reindex sources on which all hypotheses agree with f; K now counts only the remaining sources. If there are none, $M_\infty = 0$ and no probability budget is needed. Otherwise,

$$\mathbb{E}M_\infty \leq \sum_{i=1}^{K}\{1 + (h+1)(1+\log(1/\varepsilon_i))\}.$$

For a high-probability bound set

$$J_i = 1 + \lceil \log_2(1/\varepsilon_i) \rceil, \quad L = \sum_i J_i, \quad a = h + \log(2L/\delta), \quad B_\delta = (L+K)a + 2K.$$

Then, with probability at least $1-\delta$,

$$M_\infty \le (e-1)B_\delta + \log(2/\delta).$$

On a finite support with smallest positive atom p_i*, one may replace* ε_i *by* p_i*. In particular, sources uniform on arbitrary subsets of* $\{0,1\}^d$ *give an expectation of* $O(K(h+1)(d+1))$ *even with unlimited adaptive switching among disjoint regions.*

The source need not be observed by the learner. In particular the theorem covers arbitrary predictable mixture weights over the fixed D_i: choose a latent source from the current weights and then draw from that source. There is no bound on the number of switches or on the distance between the source distributions.

Proof. Couple source i to its own iid stream and let $R_{i,n}$ be the maximum D_i-risk of a hypothesis surviving the first n samples of that stream. The same union bound gives $\mathbb{P}(R_{i,n} > u) \le \min\{1, e^{h-nu}\}$. The variable $R_{i,n}$ is either zero or at least ε_i. For $u > 0$, monotonicity of the integrand gives

$$\sum_{n\ge 1} \min\{1, e^{h-nu}\} \le \int_0^\infty \min\{1, e^{h-tu}\}\, dt = (h+1)/u.$$

Consequently Tonelli's theorem, using ε_i in place of u for $0 < u < \varepsilon_i$, yields

$$\sum_{n\ge 1} \mathbb{E}R_{i,n} \le (h+1)\left\{\int_0^{\varepsilon_i} \frac{du}{\varepsilon_i} + \int_{\varepsilon_i}^1 \frac{du}{u}\right\} = (h+1)(1+\log(1/\varepsilon_i)).$$

The initial local round has risk at most one. On every visit to source i after n samples, the current consistent predictor has risk at most $R_{i,n}$, regardless of observations from other sources. Each local count is used at most once. Summing these bounds proves the expectation claim.

For high probability, for each $1 \le j \le J_i$ put $n_{i,j} = \lceil a2^j \rceil$. A union bound gives, simultaneously,

$$R_{i,n_{i,j}} \le 2^{-j}$$

except with probability $Le^{h-a} = \delta/2$. Since $2^{-J_i} < \varepsilon_i$, the last inequality implies $R_{i,n_{i,J_i}} = 0$. By monotonicity of $R_{i,n}$, the initial block costs at most $2a+1$ and each subsequent block costs at most $a + 2^{-j}$, so

$$\sum_{n\ge 0} R_{i,n} \le (J_i+1)a + 2.$$

Stop immediately before using a prefix which violates the applicable block bound. Until stopping, the conditional error sum is at most B_δ. The same exponential-supermartingale bound and the good-event union bound establish the stated infinite-horizon probability guarantee. □

Remark 12 (Bounded arbitrary departures)**.** *Either environment theorem remains valid with at most* A *extra arbitrary input rounds, with an additive* A *in the actual mistake bound and its expectation. The extra noiseless observations only shrink the version space. The clean source counters and their latent orders are unchanged. This includes a finite adversarial excursion followed by arbitrarily many further source switches.*

5.1 Randomizing within Hamming layers restores a polynomial bound

For circuits over all two-input Boolean gates, or over the AND/OR/NOT basis, a simple randomization of the input order changes the same example. Let $f(x) = \bigwedge_{i=1}^{d} x_i$, present 1^d and 0^d first, and then present the remaining Hamming levels in increasing order. Independently for each level, use a uniformly random permutation of that level. The learner remains deterministic and chooses a smallest consistent circuit using any rule depending only on the preceding history.

For $k \geq 2$, after all smaller levels have been labeled zero, the globally minimum consistent functions are precisely the surviving k-variable conjunctions. Here is the elementary reason. If h depends on the variables in R, then $h(1_R) = h(1^d) = 1$, so $|R| \geq k$. A circuit with r binary gates depends on at most $r + 1$ variables. Surviving k-conjunctions attain size $k - 1$; at that minimum size, h depends on exactly k variables. Every non-all-ones assignment on these variables has a zero extension of weight below k, so h is their conjunction. The same argument applies to AND/OR/NOT circuits with all gates charged, since unary gates add no dependencies.

Suppose that m weight-k inputs remain. The learner's chosen conjunction is wrong at precisely one of them, its own support indicator. Conditional on the complete preceding history, the next input is uniform over the m remaining inputs. Its conditional mistake probability is therefore exactly $1/m$, regardless of the history-dependent deterministic tie rule. Consequently stage k has exactly

$$H_{\binom{d}{k}} = \sum_{m=1}^{\binom{d}{k}} \frac{1}{m}$$

expected mistakes. The first two inputs and the weight-one level contribute between zero and $d + 2$ mistakes, so

$$\sum_{k=2}^{d-1} H_{\binom{d}{k}} \;\leq\; \mathbb{E}M \;\leq\; d + 2 + \sum_{k=2}^{d-1} H_{\binom{d}{k}} \;=\; \Theta(d^2).$$

For the upper bound, $H_N \leq 1 + \ln N \leq 1 + d \ln 2$. For the lower bound, for each integer $d/4 \leq k \leq d/2$, $\binom{d}{k} \geq (d/k)^k \geq 2^{d/4}$, hence $H_{\binom{d}{k}} \geq (d/4) \ln 2$; there are $\Omega(d)$ such levels. The assertion is asymptotic in d.

Thus the order still progresses strictly from low to high Hamming weight after the initial positive example; only the order inside each level is randomized. This suffices to reduce the deterministic learner's expected mistake count from exponential to quadratic in the target's minimum gate count $d - 1$. The exact phase expectations hold for every deterministic history-only tie rule. No analogous upper bound is asserted here for pure NAND circuits, because the NAND proof uses certificates rather than a classification of every minimum consistent function.

6 Adaptive shifts with a persistent reference distribution

A deterministic choice among minimum circuits also has a polynomial all-time mistake guarantee under a useful class of non-i.i.d. processes. The condition is that the next-input distribution may adapt to the whole past, but remains comparable to one fixed reference distribution. In particular, it continues to sample every reference region often enough. The reference need not be uniform.

Let $\mathcal{H}$ be a finite class of N Boolean hypotheses containing a fixed target $h_\star$. Write $\mathcal{F}_t$ for the information available after round t, and suppose the learner chooses an $\mathcal{F}_{t-1}$-measurable $h_t \in \mathcal{H}$ that is consistent with all preceding examples. The choice may depend arbitrarily on the past, including learner randomness included in the filtration. Its circuit must be chosen before seeing the current input. Labels are always $h_\star(X_t)$. Denote the conditional law of X_t given $\mathcal{F}_{t-1}$ by P_t. The

comparison below is with respect to this same filtration, including any learner randomness revealed before the current input.

Proposition 13 (Adaptive reference comparability)**.** *Suppose that, almost surely, for every round t and measurable set A,*

$$a\mu(A) \le P_t(A) \le b\mu(A), \qquad 0 < a \le 1 \le b < \infty,$$

where μ is a fixed probability distribution. Suppose also that every positive disagreement probability $\mu\{h \ne h_\star\}$, for $h \in \mathcal{H}$, is at least $0 < \gamma \le 1$. Then the total number M_∞ of mistakes satisfies

$$\mathbb{E}M_\infty \le b\left[1 + \frac{\log N + 1}{a}\left(1 + \log\frac{1}{\gamma}\right)\right].$$

In particular, $M_\infty < \infty$ almost surely.

Proof. For $h \in \mathcal{H}$, put $p(h) = \mu\{h \ne h_\star\}$ and let $\mathcal{V}_n$ be the hypotheses consistent with the first n examples. The conditional lower bound implies

$$\Pr(h \in \mathcal{V}_n) \le (1 - ap(h))^n \le e^{-anp(h)}.$$

Set $R_n = \max_{h\in\mathcal{V}_n} p(h)$. By a union bound,

$$\Pr(R_n > u) \le \min\{1, Ne^{-anu}\}, \qquad u > 0.$$

For $0 < u < \gamma$, the same bound holds with u replaced by γ, because each positive $p(h)$ is at least γ. For any $q > 0$, monotonicity gives

$$\sum_{n=0}^{\infty} \min\{1, Ne^{-anq}\} \le 1 + \int_0^\infty \min\{1, Ne^{-atq}\}\,dt = 1 + \frac{\log N + 1}{aq}.$$

Integrating the tail probability over $0 < u < 1$, splitting at γ, and using Tonelli's theorem therefore yields

$$\mathbb{E}\sum_{n=0}^{\infty} R_n \le 1 + \frac{\log N + 1}{a}\left(1 + \log\frac{1}{\gamma}\right).$$

Conditional on $\mathcal{F}_{t-1}$, the learner's mistake probability is at most $b\,p(h_t) \le bR_{t-1}$. Summation proves the claim. □

For the uniform reference on $\{0,1\}^d$, one may take $\gamma = 2^{-d}$. If hypotheses have binary descriptions of at most L bits, then $N < 2^{L+1}$, giving

$$\mathbb{E}M_\infty = O\!\left(\frac{b}{a}(L+1)(d+1)\right).$$

This applies to an exact minimum-description consistent circuit, since a realizing target of description length L keeps every selected circuit within that class. For minimum gate count s, standard circuit counting gives $\log N = O((s+1)\log(d+s+2))$. No efficient algorithm for finding the exact minimum is asserted. For a nonuniform finite reference, γ may instead be bounded below by its smallest positive atom; polynomial bounds hold when $\log(1/\gamma)$ and b/a are polynomial in the problem parameters.

For example, a process of the form

$$P_t = (1-\eta)\mu + \eta Q_t, \qquad 0 \le \eta < 1, \qquad Q_t(A) \le B\mu(A),$$

may choose Q_t adaptively. It satisfies the proposition with $a = 1 - \eta$ and $b = 1 - \eta + \eta B$. It need never become i.i.d. or converge to a limiting distribution.

Summably many departures from the comparison condition. The same conclusion has a robust extension. Suppose there are nonnegative $\mathcal{F}_{t-1}$-measurable ε_t such that

$$a\mu(A) - \varepsilon_t \le P_t(A) \le b\mu(A) + \varepsilon_t, \qquad \sum_{t\ge 1} \varepsilon_t \le V \quad \text{almost surely.}$$

Then

$$\mathbb{E}M_\infty \le V + b\left[1 + \frac{\log N + V + 1}{a}\left(1 + \log\frac{1}{\gamma}\right)\right].$$

Indeed, for each fixed h, the process

$$\mathbf{1}\{h \in \mathcal{V}_n\} \exp\left(anp(h) - \sum_{t=1}^{n} \varepsilon_t\right)$$

is a nonnegative supermartingale. Thus $\Pr(h \in \mathcal{V}_n) \le e^{V - anp(h)}$. The preceding proof applies with N replaced by Ne^V, while the conditional mistake probability is at most $bR_{t-1} + \varepsilon_t$. In particular, with $a = b = 1$ this covers $\sum_t \|P_t - \mu\|_{\mathrm{TV}} \le V$. Here total variation is $\sup_A |P(A) - Q(A)|$. This assumption controls cumulative distance from a fixed reference; it is different from bounding only the successive changes $\sum_t \|P_{t+1} - P_t\|_{\mathrm{TV}}$.

6.1 Tiny persistent shifts can defeat deterministic minimum circuits

An upper density bound alone does not suffice even for exact minimum circuits. The failure can persist when every conditional input law is exponentially close to uniform and, for a fixed ranking of circuits, the sum of successive distribution changes is exponentially small. This construction uses repeated inputs.

Proposition 14 (Upper smoothness does not ensure few mistakes)**.** *Use a fixed finite fan-in-two gate basis containing AND, and count gates. Let $d \ge 4$ be divisible by four, let μ be uniform on $\{0,1\}^d$, and put $\delta = e^{-d/8}$. For every deterministic, history-only exact-minimum consistent circuit rule, there is an adaptive input process with target $f(x) = \bigwedge_{j=1}^{d} x_j$, of minimum size $d-1$, such that*

$$P_t(A) \le \frac{\mu(A)}{1-\delta} \quad \textit{for every set } A, \qquad \|P_t - \mu\|_{\mathrm{TV}} \le \delta \quad \textit{for every round } t,$$

yet the learner makes at least $2^{d/2}$ mistakes almost surely. If the rule selects the first consistent minimum circuit in a fixed ranking, the same process also satisfies

$$\sum_{t\ge 1} \|P_{t+1} - P_t\|_{\mathrm{TV}} \le \frac{\delta}{1-\delta} \quad \textit{almost surely.}$$

There is a deterministic horizon T with $\log T = O(d)$ at which $\mathbb{E}M_T \ge 2^{d/2-1}$.

Proof. Put $k = 3d/4$, let $R = \{x : |x| < k\}$, and write $r = |R|$, $m = \binom{d}{k}$, and $N = 2^d$. First draw independently and uniformly from $B = R \cup \{1^d\}$ until every point of B has appeared. This stopping time is finite almost surely. These examples force any consistent function to depend on at least k coordinates: a function depending on a set S of fewer than k coordinates cannot distinguish 1^d from $1_S \in R$. A circuit with s binary gates depends on at most $s+1$ coordinates. Every k-variable conjunction is consistent and has $k-1$ gates, so minimum size is exactly $k-1$.

Moreover, every minimum circuit computes a k-variable conjunction. It depends on exactly k coordinates, say S, and is one on their all-ones assignment. Every other assignment to S occurs on a point of weight below k and is labeled zero. Its function is therefore precisely $\bigwedge_{j\in S} x_j$.

While any k-conjunction remains consistent, let S_t be the support of the learner's current minimum circuit and choose P_t uniform on $B \cup \{1_{S_t}\}$. The current circuit is determined by the past, so this defines a valid adaptive input process. A draw from B causes no mistake and eliminates no remaining k-conjunction. A draw of 1_{S_t} has conditional probability $1/(r+2)$, causes a mistake, and eliminates exactly that conjunction: two distinct k-subsets are not contained in one another. Therefore minimum size remains $k-1$ until all m conjunctions have been eliminated. Those m errors occur almost surely. Thereafter retain the final distribution, whose entire support has already been observed.

Hoeffding's inequality gives

$$1 - r/N = \Pr\{\mathrm{Bin}(d, 1/2) \geq 3d/4\} \leq e^{-d/8} = \delta.$$

Every conditional law is uniform on a set of size $r+1$ or $r+2$. Its density relative to μ is at most $N/(r+1) \leq (1-\delta)^{-1}$, and its total variation distance from μ is at most $1-(r+1)/N \leq \delta$, including the initial and final stages. Under a fixed ranking, repeated points of B leave the selected circuit unchanged. There are m hard phases. The initial switch and the $m-1$ switches between their laws each have total variation $1/(r+2)$; there is no final switch. Thus total successive variation is $m/(r+2) \leq \delta/(1-\delta)$, using $m \leq N - r \leq N\delta$.

Finally, $m = \binom{d}{d/4} \geq (d/(d/4))^{d/4} = 2^{d/2}$. This construction permits repeated inputs; the earlier exponential lower bounds use a permutation with no repeats. The expected time through the initial stage and all hard phases is

$$(r+1)H_{r+1} + m(r+2),$$

where H_n is the nth harmonic number. Taking T to be the ceiling of twice this expectation gives $\mathbb{E}M_T \geq m/2$ by Markov's inequality, and $\log T = O(d)$. This horizon is exponential in d; only its logarithm is linear in d. □

7 A precise genericity condition

This final result applies to a particular simplest-consistent selector with no prescribed tie-breaking, but its assumption is target-relative and must be justified separately for the application. Let V_t be all surviving descriptions no longer than the target description, let N be their initial count, and let q_t be the fraction of V_t wrong on the current input. The target survives, so $q_t < 1$ and, pathwise,

$$\sum_t q_t \leq \sum_t -\log(1-q_t) \leq \log N.$$

Let Z_t indicate a mistake by the selected simplest description.

Theorem 15 (Conditional genericity)**.** *Suppose, for a constant $0 < c \leq 1$, on every round with positive conditional mistake probability,*

$$\mathbb{E}[q_t \mid Z_t = 1, \mathcal{F}_{t-1}] \geq c.$$

Equivalently, without conditioning on null events, assume $\mathbb{E}[Z_t q_t \mid \mathcal{F}_{t-1}] \geq c\mathbb{E}[Z_t \mid \mathcal{F}_{t-1}]$. Then

$$\mathbb{E}M_\infty \leq \frac{\log N}{c}, \qquad \mathbb{P}\left\{M_\infty \geq \frac{\log N + \log(1/\delta)}{c(1-e^{-1})}\right\} \leq \delta.$$

It suffices that $\mathbb{P}(q_t \geq \alpha \mid Z_t = 1, \mathcal{F}_{t-1}) \geq \beta$ with $c = \alpha\beta$. No independence, stationarity, or iid input assumption is needed.

Proof. The expectation follows by summing $c\mathbb{E}Z_t \le \mathbb{E}[Z_t q_t] \le \mathbb{E}q_t$. For the tail let $\theta = -\log(1 - c(1-e^{-1})) \ge c(1-e^{-1})$. Convexity on $[0,1]$ gives $e^{-q} \le 1-(1-e^{-1})q$. Thus

$$\mathbb{E}[e^{\theta Z_t - q_t} \mid \mathcal{F}_{t-1}] \le 1 - p_t + e^{\theta}\{p_t - (1-e^{-1})\mathbb{E}[Z_t q_t \mid \mathcal{F}_{t-1}]\} \le 1,$$

where $p_t = \mathbb{E}[Z_t \mid \mathcal{F}_{t-1}]$. Hence $\exp\{\theta M_T - \sum_{t\le T} q_t\}$ is a nonnegative supermartingale. Since $\sum_t q_t \le \log N$, Ville's inequality proves the result, uniformly over all finite horizons and therefore at infinity. □

Interpretation. These are three distinct positive routes. Fresh uniform minimum-tier sampling supplies a theorem for arbitrary adaptive inputs by changing the tie rule. Interleaved environments give an explicit structured input class for every history-dependent tie rule, with a logarithmic domain-size overhead. Conditional genericity directly formalizes the proposed constant-fraction-elimination intuition. The environment theorems do not assert that a constant fraction is removed on each mistake; they prove a polynomial total by a different survival argument. None of these claims by itself establishes arbitrary-order guarantees for a specified Gaussian neural posterior.

8 What follows, and what remains open

The exact-minimality issue in the AND example is resolved for both unrestricted binary-gate circuits and NAND-only circuits. No difficult general circuit lower bound is needed: support size proves the former, and a small positive certificate proves the latter. The target labels are fixed in advance. For each deterministic history-based learner, the adversarial construction can be simulated in advance into one fixed permutation; it does not assert one permutation that defeats all learners.

There are rigorous positive results without changing that deterministic selection rule. Independent shuffling inside each Hamming-weight level already changes the conjunction example from exponential to quadratic expected mistakes in the binary-gate basis. The environment theorems allow unrelated and disjoint regions, arbitrary adaptive switches and mixtures, and infinitely many rounds. Their polynomial bound charges the number of fixed environments and their smallest positive disagreement probabilities. Persistent-reference bounds permit continuing adaptive drift instead. By contrast, tiny but persistent omissions of carefully selected rare inputs can still produce exponentially many mistakes, even with exponentially small distance from uniform and total successive drift. These are explicit sufficient conditions and obstructions, not a measure under which most conceivable input sequences are typical.

The genericity premise captures a different intuition: errors of the chosen simplest rule should often also be errors of many surviving simple rules. Its potential argument is exact, but verifying the premise for realistic deterministic data streams remains a substantive task. The randomized minimum-size theorem needs no such premise; it pays a separate counting budget at each complexity level. Ordinary Bayesian prediction under a fixed circuit-description prior has yet another potential: cumulative log loss is bounded by the target's negative log prior mass. These mechanisms should not be identified with each other.

Nothing here establishes an arbitrary-order polynomial mistake bound for the particular Gaussian neural-network learners studied elsewhere. Nor does the exponential circuit-Occam example disprove such a bound for their posterior mixtures. Transferring a result requires controlling the actual prediction law, not merely the complexity of one selected mode.

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